

EX.NO:3 INHERITANCE PROGRAM FOR AN EMPLOYEE CLASS

```
import java.util.*;
class Employee {
    String emp_name;
    int emp_id;
    String address;
    String mail_id;
    String mob_number;

    Employee(String emp_name, int emp_id, String address, String mail_id, String
mob_number) {
        this.emp_name = emp_name;
        this.emp_id = emp_id;
        this.address = address;
        this.mail_id = mail_id;
        this.mob_number = mob_number;
    }

    void display() {
        System.out.println("Employee Name : " + emp_name);
        System.out.println("Employee ID : " + emp_id);
        System.out.println("Address : " + address);
        System.out.println("Mail ID : " + mail_id);
        System.out.println("Mobile Number : " + mob_number);
    }
}

class Programmer extends Employee {
    double basicpay, da, hra, pf, fund, grosssalary, netsalary;

    Programmer(String emp_name, int emp_id, String address, String mail_id, String
mob_number, double bp) {
        super(emp_name, emp_id, address, mail_id, mob_number);
        basicpay = bp;
    }

    void getPaySlip() {
        da = basicpay * 97 / 100;
        hra = basicpay * 10 / 100;
        pf = basicpay * 12 / 100;
        fund = basicpay * 0.1 / 100;

        grosssalary = basicpay + da + hra + pf + fund;
        netsalary = grosssalary - pf - fund;
    }

    void disp() {
        System.out.println("\n***** PROGRAMMER PAY SLIP *****");
        display();
        System.out.println("Basic Pay : " + basicpay);
        System.out.println("Gross Salary : " + grosssalary);
    }
}
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        System.out.println("Net Salary  : " + netsalary);
    }
}

class AssistantProfessor extends Employee {
    double basicpay, da, hra, pf, fund, grosssalary, netsalary;

    AssistantProfessor(String emp_name, int emp_id, String address, String mail_id, String
mob_number, double bp) {
        super(emp_name, emp_id, address, mail_id, mob_number);
        basicpay = bp;
    }

    void getPaySlip() {
        da = basicpay * 97 / 100;
        hra = basicpay * 10 / 100;
        pf = basicpay * 12 / 100;
        fund = basicpay * 0.1 / 100;

        grosssalary = basicpay + da + hra + pf + fund;
        netsalary = grosssalary - pf - fund;
    }

    void disp() {
        System.out.println("\n***** ASSISTANT PROFESSOR PAY SLIP *****");
        display();
        System.out.println("Basic Pay   : " + basicpay);
        System.out.println("Gross Salary : " + grosssalary);
        System.out.println("Net Salary  : " + netsalary);
    }
}

class AssociateProfessor extends Employee {
    double basicpay, da, hra, pf, fund, grosssalary, netsalary;

    AssociateProfessor(String emp_name, int emp_id, String address, String mail_id, String
mob_number, double bp) {
        super(emp_name, emp_id, address, mail_id, mob_number);
        basicpay = bp;
    }

    void getPaySlip() {
        da = basicpay * 97 / 100;
        hra = basicpay * 10 / 100;
        pf = basicpay * 12 / 100;
        fund = basicpay * 0.1 / 100;

        grosssalary = basicpay + da + hra + pf + fund;
        netsalary = grosssalary - pf - fund;
    }

    void disp() {

```

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        System.out.println("\n***** ASSOCIATE PROFESSOR PAY SLIP *****");
        display();
        System.out.println("Basic Pay   : " + basicpay);
        System.out.println("Gross Salary : " + grosssalary);
        System.out.println("Net Salary  : " + netsalary);
    }
}

class Professor extends Employee {
    double basicpay, da, hra, pf, fund, grosssalary, netsalary;

    Professor(String emp_name, int emp_id, String address, String mail_id, String mob_number,
double bp) {
        super(emp_name, emp_id, address, mail_id, mob_number);
        basicpay = bp;
    }

    void getPaySlip() {
        da = basicpay * 97 / 100;
        hra = basicpay * 10 / 100;
        pf = basicpay * 12 / 100;
        fund = basicpay * 0.1 / 100;

        grosssalary = basicpay + da + hra + pf + fund;
        netsalary = grosssalary - pf - fund;
    }

    void disp() {
        System.out.println("\n***** PROFESSOR PAY SLIP *****");
        display();
        System.out.println("Basic Pay   : " + basicpay);
        System.out.println("Gross Salary : " + grosssalary);
        System.out.println("Net Salary  : " + netsalary);
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner in = new Scanner(System.in);

        System.out.print("Enter Employee Name: ");
        String name = in.nextLine();

        System.out.print("Enter Employee ID: ");
        int id = in.nextInt();
        in.nextLine();

        System.out.print("Enter Address: ");
        String add = in.nextLine();

        System.out.print("Enter Mail ID: ");

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String mail = in.nextLine();

System.out.print("Enter Mobile Number: ");
String mob = in.nextLine();

System.out.print("Enter Basic Pay: ");
double bp = in.nextDouble();

System.out.println("\nChoose Designation");
System.out.println("1. Programmer");
System.out.println("2. Assistant Professor");
System.out.println("3. Associate Professor");
System.out.println("4. Professor");
System.out.println("5. Exit");

System.out.print("Enter Choice: ");
int desg = in.nextInt();

switch (desg) {
    case 1:
        Programmer p = new Programmer(name, id, add, mail, mob, bp);
        p.getPaySlip();
        p.disp();
        break;
    case 2:
        AssistantProfessor ap = new AssistantProfessor(name, id, add, mail, mob, bp);
        ap.getPaySlip();
        ap.disp();
        break;
    case 3:
        AssociateProfessor asp = new AssociateProfessor(name, id, add, mail, mob, bp);
        asp.getPaySlip();
        asp.disp();
        break;
    case 4:
        Professor pr = new Professor(name, id, add, mail, mob, bp);
        pr.getPaySlip();
        pr.disp();
        break;
    case 5:
        System.out.println("Program Exited.");
        break;
    default:
        System.out.println("Invalid Choice!");
}

in.close();
}
```

OUTPUT:

Enter Employee Name: Arun
Enter Employee ID: 201
Enter Address: Chennai
Enter Mail ID: arun@gmail.com
Enter Mobile Number: 9876543210
Enter Basic Pay: 45000

Choose Designation

1. Programmer
2. Assistant Professor
3. Associate Professor
4. Professor
5. Exit

Enter Choice: 2

***** ASSISTANT PROFESSOR PAY SLIP *****

Employee Name : Arun
Employee ID : 201
Address : Chennai
Mail ID : arun@gmail.com
Mobile Number : 9876543210
Basic Pay : 45000.0
Gross Salary : 98595.0
Net Salary : 93150.0

EX.NO:4 CREATE AN ABSTRACT CLASS – SHAPE CLASS

PROGRAM:

```
import java.util.*;
abstract class Shape {
    public int x, y;
    public abstract void printArea();
}
class Rectangle1 extends Shape {
    public void printArea() {
        float area;
        area = x * y;
        System.out.println("Area of Rectangle is " + area);
    }
}
class Triangle extends Shape {
    public void printArea() {
        float area;
        area = (x * y) / 2.0f;
        System.out.println("Area of Triangle is " + area);
    }
}
class Circle extends Shape {
    public void printArea() {
        float area;
        area = (22.0f * x * x) / 7.0f;
        System.out.println("Area of Circle is " + area);
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Length and Breadth: ");
        int x1 = sc.nextInt();
        int y1 = sc.nextInt();
        Rectangle1 r = new Rectangle1();
        r.x = x1;
        r.y = y1;
        r.printArea();
        Triangle t = new Triangle();
        t.x = x1;
        t.y = y1;
        t.printArea();
        Circle c = new Circle();
        c.x = x1;
        c.printArea();
        sc.close();
    }
}
```

OUTPUT:

Enter Length and Breadth: 10

5

Area of Rectangle is 50.0

Area of Triangle is 25.0

Area of Circle is 314.2857